

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Returns, plant and tax depreciation

250 kWp
PLANT CAPACITY

APSPDCL · COMMERCIAL ROOFTOP

250 KWP GRID-TIED SOLAR PLANT

Net investment ₹1.02 Cr after input tax credit, generating 3,83,250 units in year one against an energy charge of ₹7.65 per unit.

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

PAYBACK

3.9 yrs

Post-tax, undiscounted

PROJECT IRR

26.2%

Unlevered, 30 years

COST OF SOLAR ENERGY

₹3.32

per kWh, against ₹7.65 from the grid

NET PRESENT VALUE

₹1.73 Cr

Discounted at 10%

PLANT AND GENERATION

Plant capacity	250 kWp
Year 1 generation	3,83,250 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹27.44 L

INVESTMENT AND TAX BASIS

Price per watt-peak	₹43.00 / Wp (incl. GST)
Project price incl. GST	₹1.07 Cr
GST at 5% — recovered as ITC	₹5.12 L
Net capitalised cost	₹1.02 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹25.77 L

ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	1,02,38,095	40,95,238	61,42,857	10,30,689
2	61,42,857	24,57,143	36,85,714	6,18,414
3	36,85,715	14,74,286	22,11,429	3,71,048
4	22,11,428	8,84,571	13,26,857	2,22,629
5	13,26,857	5,30,743	7,96,114	1,33,577
6	7,96,115	3,18,446	4,77,669	80,146
7	4,77,668	1,91,067	2,86,601	48,088
8	2,86,601	1,14,640	1,71,961	28,853
9	1,71,960	68,784	1,03,176	17,312
10	1,03,177	41,271	61,906	10,387
11	61,906	24,762	37,144	6,232
12	37,143	14,857	22,286	3,739
Tax shield, years 1-12				25,71,115

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

250 kWp

PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	3,83,250	27,44,070	—	40,95,238	(3,40,062)	30,84,132	30,84,132
2	3,81,334	28,39,564	1,50,000	24,57,143	58,496	26,31,068	57,15,200
3	3,79,427	29,38,380	1,50,000	14,74,286	3,30,731	24,57,649	81,72,849
4	3,77,530	30,40,636	1,50,000	8,84,571	5,04,886	23,85,750	1,05,58,599
5	3,75,642	31,46,450	1,50,000	5,30,743	6,20,569	23,75,881	1,29,34,480
6	3,73,764	32,55,947	1,50,000	3,18,446	7,01,558	24,04,388	1,53,38,868
7	3,71,895	33,69,254	1,50,000	1,91,067	7,62,134	24,57,120	1,77,95,988
8	3,70,036	34,86,504	1,50,000	1,14,640	8,10,879	25,25,625	2,03,21,613
9	3,68,186	36,07,834	1,50,000	68,784	8,52,956	26,04,878	2,29,26,491
10	3,66,345	37,33,387	1,50,000	41,271	8,91,480	26,91,907	2,56,18,398
11	3,64,513	38,63,308	1,50,000	24,762	9,28,333	27,84,975	2,84,03,373
12	3,62,690	39,97,752	1,50,000	14,857	9,64,663	28,83,089	3,12,86,462
13	3,60,877	41,36,873	1,50,000	8,914	10,01,173	29,85,701	3,42,72,163
14	3,59,073	42,80,837	1,50,000	5,349	10,38,303	30,92,534	3,73,64,697
15	3,57,277	44,29,810	1,50,000	3,209	10,76,335	32,03,475	4,05,68,172
16	3,55,491	45,83,967	1,50,000	1,926	11,15,456	33,18,511	4,38,86,683
17	3,53,713	47,43,489	1,50,000	1,155	11,55,799	34,37,691	4,73,24,374
18	3,51,945	49,08,563	1,50,000	693	11,97,461	35,61,102	5,08,85,476
19	3,50,185	50,79,380	1,50,000	416	12,40,522	36,88,859	5,45,74,335
20	3,48,434	52,56,143	1,50,000	250	12,85,051	38,21,092	5,83,95,427
21	3,46,692	54,39,057	1,50,000	150	13,31,112	39,57,945	6,23,53,372
22	3,44,959	56,28,336	1,50,000	90	13,78,765	40,99,571	6,64,52,943
23	3,43,234	58,24,202	1,50,000	54	14,28,070	42,46,132	7,06,99,075
24	3,41,518	60,26,884	1,50,000	32	14,79,086	43,97,798	7,50,96,873
25	3,39,810	62,36,620	1,50,000	19	15,31,876	45,54,744	7,96,51,617
26	3,38,111	64,53,654	1,50,000	12	15,86,501	47,17,153	8,43,68,770
27	3,36,420	66,78,241	1,50,000	7	16,43,026	48,85,215	8,92,53,985
28	3,34,738	69,10,644	1,50,000	4	17,01,518	50,59,126	9,43,13,111
29	3,33,065	71,51,135	1,50,000	3	17,62,045	52,39,090	9,95,52,201
30	3,31,399	73,99,994	1,50,000	2	18,24,678	54,25,316	10,49,77,517
Total	1,07,01,553	14,11,90,915	43,50,000	1,02,38,093	3,18,63,400	10,49,77,517	10,49,77,517

RETURN METRICS

Simple payback	3.9 yrs
Discounted payback	5.2 yrs
Project IRR, unlevered	26.2%
Net present value at 10%	₹1.73 Cr

OVER THE FULL TERM

Levelised cost of energy	₹3.32 / kWh
Cumulative return, 30 years	925%
Total net cash flow, after tax	₹10.50 Cr
30-year generation	1,07,01,552 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹3.32
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹3.32 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 1,07,01,552 units, avoiding 8,775.3 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹1.07 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(iia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 250 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.